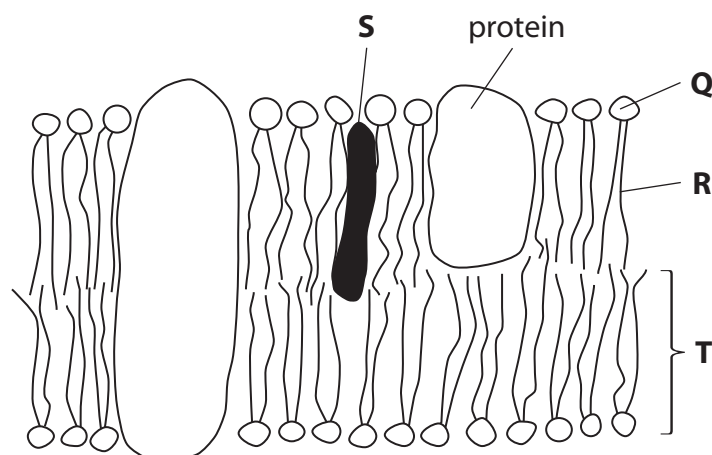


- 8 The structure of the cell membrane affects the properties of the cell membrane.

The diagram shows the structure of part of a cell membrane.



- (a) (i) The magnification of this diagram is 1×10^7 .

The structure labelled **T** is 2 units long.

What is this unit?

(1)

- ☐ **A** cm
- ☐ **B** mm
- ☐ **C** μm
- ☐ **D** nm

- (ii) Which structures contain a phosphate group?

(1)

- ☐ **A** **Q** and **R**
- ☐ **B** **Q** and **T**
- ☐ **C** **R** and **S**
- ☐ **D** **S** and **T**

- (iii) Which ratio of the structures affects the fluidity of this membrane?

(1)

- ☐ **A** **Q**:**T**
- ☐ **B** **R**:**Q**
- ☐ **C** **R**:**T**
- ☐ **D** **S**:**T**

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(b) Explain the role of the primary structure in determining the properties of the protein labelled in the diagram.

(4)

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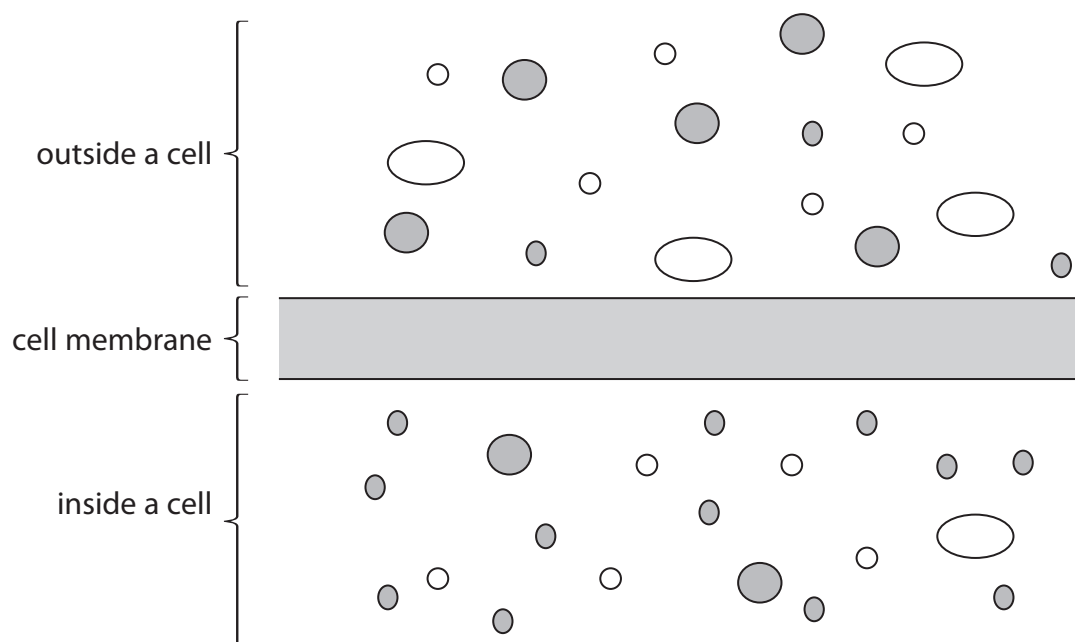
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P 6 5 8 1 2 A 0 2 7 3 2

*(c) The diagram and table give information about some molecules found inside and outside a cell.

Molecule	Key	Description
E	○	small dipolar molecule consisting of two hydrogen atoms and one oxygen atom
F	●	large polar molecule
G	○	non-polar molecule
H	●	small polar molecule



Explain why each of these molecules **enters** the cell by a different mechanism.

Use the information in the table and the diagram to support your answer.

(6)

(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 80 MARKS

